

Statistics II: Introduction to Inference

Module 1: Introduction and Recaps

February 11, 2026

The job of a statistician is to **infer** about an unknown **population** from a representative subset of the population, called **samples**. We need to understand the three highlighted terms first.

1 Population

Population is a collection of units or measurements. In statistical inference we seek information about the population. A population can be finite or infinite. Let us denote the population by $\mathbb{P}$.

Some Examples.

- (i) Suppose you are interested in the average marks obtained by BSDS students in Statistics II. Then the population $\mathbb{P}_1$ is the class of marks in Statistics II of all students who took this course in the past year, this year and will be taking this course in future. The quantity of interest is the average of $\mathbb{P}_1$.
- (ii) Suppose you are interested in the average household size for the residents in Delhi in year 2025. Then the population $\mathbb{P}_2$ is the class of all the household sizes in Delhi in 2025. The quantity of interest is the average of $\mathbb{P}_2$.
- (iii) Suppose you are interested in the variation in battery-life of Samsung mobiles of a particular model A. Then the population $\mathbb{P}_3$ is the collection of battery-life of all Samsung mobiles of model A. The quantity of interest is the variance (or, an appropriate measure of scatterness) of $\mathbb{P}_3$.

From the examples, it is clear that the population $\mathbb{P}$ is nothing but **a huge collection of measurements/numbers**. These numbers (if were known) can be expressed using a histogram (when continuous) or bar-diagram (when discrete), where x -axis represents the values of $\mathbb{P}$ and y -axis represents the relative frequency [frequency of an interval (for histogram), or a number (for bar-diagram) divided by the population size].

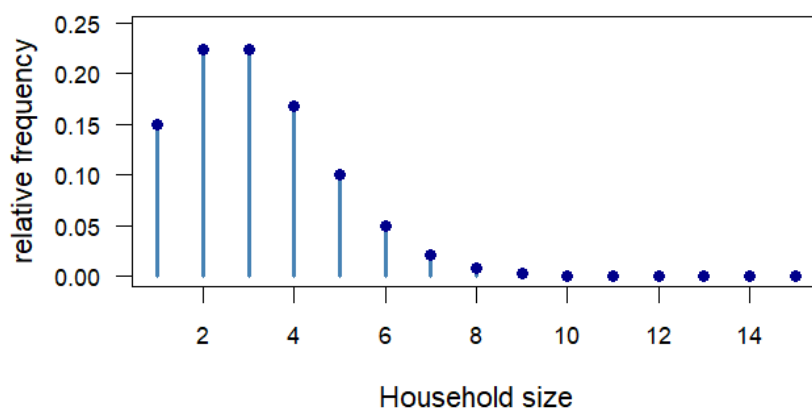


Figure 1: Relative frequency plot of household sizes in Delhi in 2025.

When the population is of discrete nature (e.g., $\mathbb{P}_2$: the household size of Delhi), then the corresponding bar-diagram of relative frequencies (see, e.g., Figure 1) can be equivalently represented by a function f with domain as the values of $\mathbb{P}_2$ and range in $[0, 1]$, and satisfying that the sum of non-zero f values is 1 (being sum of relative frequencies), i.e., $\sum_{x: f(x) > 0} f(x) = 1$. Now, observe that the above-mentioned properties of f are satisfied by a probability mass function (p.m.f.). So, when discrete, the population can be represented by a p.m.f.

When the population is of continuous type (e.g., $\mathbb{P}_3$: battery-life of Samsung mobiles of type A) , then due to its huge size, it is possible to choose extremely finer bins of the histogram (observe that, histogram with finer bins provide more information about the population than thicker bins). Further, the nearby bins are likely to have similar relative frequencies. So, the relative frequencies gradually increase or decline. It is therefore possible to (approximately) represent the population by a function g with the domain as the values of $\mathbb{P}_3$ and range in $\mathbb{R}$, which provides the limiting relative frequency as the bin size decreases to almost zero, and it satisfies $\int_0^\infty g(x)dx = 1$ (total relative frequency). A graphical representation of the histogram and the corresponding g is given in Figure 2.

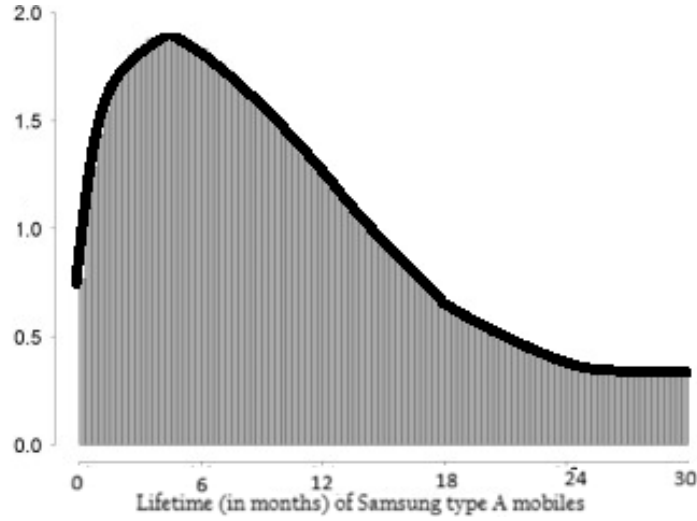


Figure 2: Relative frequency plot of household sizes in Delhi in 2025.

Now, observe that the above-mentioned properties of g are satisfied by a probability density function (p.d.f.). So, when continuous, the population can be represented by a p.d.f.

Therefore, a population can be completely characterized by the corresponding p.m.f. or p.d.f., and equivalently from a c.d.f. (as the p.m.f. / p.d.f. can be derived from c.d.f.). For e.g., $N(0, 1)$ is a valid description of a population.

Remarks.

- R1. There is *no randomness* in the population. The function p.m.f./p.d.f. representing the population is completely deterministic.
- R2. If the p.d.f./p.m.f. representing the population is completely known then all the questions related to the population (e.g., average, variance, proportions, etc.) can also be derived.

2 Random Sample

As the population $\mathbb{P}$ usually huge, it can not be enumerated entirely. Instead a small, representative subset of the population is considered to infer about the population. This representative subset (of size n , say) is called a sample, and denoted by $X_1, \dots, X_n$.

Consider the first sample X_1 . Before, observing X_1 , one can't predict it's value exactly. However, one can expect that X_1 can take a value within the range of $\mathbb{P}$, and further, every value in the range of $\mathbb{P}$ is not equally likely for X_1 . For example, a sample from $\mathbb{P}_2$ (household size) is much more likely

to take a value below 6 (see Figure 1) than above that. Similarly, a sample from $\mathbb{P}_3$, is much more likely to take a below 12 months than above 24 months. In fact the best guess is to assign the following probability to X_1 :

$$P(a < X_1 \leq b) = \int_a^b g(x)dx,$$

when $\mathbb{P}$ is continuous type, and g is the representative p.d.f. of $\mathbb{P}$; or,

$$P(X_1 = x) = f(x),$$

when $\mathbb{P}$ is of discrete type, and f is the representative p.m.f. of $\mathbb{P}$.

In other words, X_1 is a random variable with the probability distribution characterized by the p.d.f. g , or the p.m.f. f , representing the population. The same can be said about $X_2, \dots, X_n$. Therefore, $X_1, \dots, X_n$ are identically distributed random variables with the probability distribution characterized by the p.d.f. g , or the p.m.f. f , representing the population.

Finally, if the samples are randomly taken from the population, then we can also assume that $X_1, \dots, X_n$ are independently distributed.

Combining these two facts, we say $X_1, \dots, X_n$ are independent and identically distributed (i.i.d.) random variables with the probability distribution characterized by the p.d.f. g , or the p.m.f. f , representing the population. Notationally, we write $X_i \stackrel{i.i.d.}{\sim} g$ or $X_i \stackrel{i.i.d.}{\sim} f$ for $i = 1, \dots, n$.

3 Statistical Model and Inference

The population $\mathbb{P}$ is usually NOT known. Consequently, the p.d.f./p.m.f. representing $\mathbb{P}$ is also unknown. The job of a statistician is to *infer* about $\mathbb{P}$ from the random sample(s).

However, it is difficult to derive the entire population based on a few samples. Therefore, based on the domain knowledge, a statistician usually *guesses* the shape of the p.d.f./p.m.f. associated with $\mathbb{P}$, without pin-pointing it exactly. For example, one may guess that the distribution of income of Indian households in Delhi in year 2025 can be represented by a $\text{Gamma}(\alpha, \beta)$ distribution. Here a family of distributions is specified as Gamma, but the exact candidate is not specified (as (α, β) remains unspecified).

Remarks.

- R1. The art of deriving the shape of the p.d.f./p.m.f. is called *modeling*. The assumed form (with some unknown constants) is called a *parametric model*, and unknown constants are called *parameters*. In the household income example, (α, β) is the vector of parameters.
- R2. Note that, given the model (for, e.g., given that household income is distributed as $\text{Gamma}(\alpha, \beta)$), it is enough to estimate the parameters (here, (α, β)) using the samples in order to estimate the population characteristics. If the parameters are well estimated, then the entire population is also estimated well.
- R3. The task of *parametric inference* is to estimate the parameters, or, functions of parameters from the random sample $X_1, \dots, X_n$.

4 Statistic and Sample Distribution

Let $X_i \stackrel{i.i.d.}{\sim} F_{\theta}$ for $i = 1, \dots, n$. Here F_{θ} represents a form of a distribution with parameter vector θ . Suppose we are interested in a function of θ , say $h(\theta)$. [For example, Let $X_i \stackrel{i.i.d.}{\sim} \text{normal}(\mu, \sigma^2)$ for $i = 1, \dots, n$. Here $\theta = (\mu, \sigma^2)'$, and we are interested in the population average $\mu = g(\theta)$.]

Given the random sample $\{X_1, \dots, X_n\}$, a statistician uses a function of $\{X_1, \dots, X_n\}$, say $T_n = T(\{X_1, \dots, X_n\})$, to estimate $g(\theta)$. For instance, in the above example, $\bar{X} = (X_1 + \dots + X_n)/n$ was used to estimate $\mu = g(\theta)$. Such a function of $\{X_1, \dots, X_n\}$ is called a *statistic*.

Remarks.

- R1. The collection of all possible values of $(X_1, \dots, X_n)$ is called the **sample space**, and is usually denoted by $\mathcal{X}_n$ (n being the sample size). For example, if $X_i \stackrel{i.i.d.}{\sim} \text{normal}(\mu, \sigma^2)$ for $i = 1, \dots, n$, then the corresponding sample space is $\mathcal{X}_n = \mathbb{R} \times \dots \times \mathbb{R} = \mathbb{R}^n = \{(x_1, \dots, x_n) : x_1 \in \mathbb{R}, \dots, x_n \in \mathbb{R}\}$. A statistic T is a function from the domain $\mathcal{X}_n$ to the real line $\mathbb{R}$.
- R2. As $\{X_1, \dots, X_n\}$ is random, so is a statistic $T_n = T(X_1, \dots, X_n)$. The probability distribution of a statistic is called the **sampling distribution** of the statistic.
- R3. Note that, as the distribution of $\{X_1, \dots, X_n\}$ depends of the parameters θ , the sampling distribution of a statistic usually also depends on θ . For example, let $X_1, \dots, X_n$ be a random sample of size n from $\text{normal}(\mu, \sigma^2)$ distribution. Then $T_n = \sum_{i=1}^n X_i$ is a statistic, and its distribution is $\text{normal}(\mu, \sigma^2/n)$.

[IN THE CLASS WE HAVE STUDIED SEVERAL TECHNIQUES FOR DERIVING SAMPLING DISTRIBUTIONS. SOME PROBLEMS BASED ON THESE TECHNIQUES ARE GIVEN IN THE PROBLEM SET 1.]